

LAB EXPERIMENTS

Q15) Write a C program that can perform a letter frequency attack on an additive cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify “give me the top 10 possible plaintexts.”

Aim:

To write a C program that performs a **letter frequency attack** on an **Additive (Caesar) Cipher** by trying all possible keys and displaying the most probable plaintexts. The program allows the user to specify the number of possible plaintexts (Top N) to display.

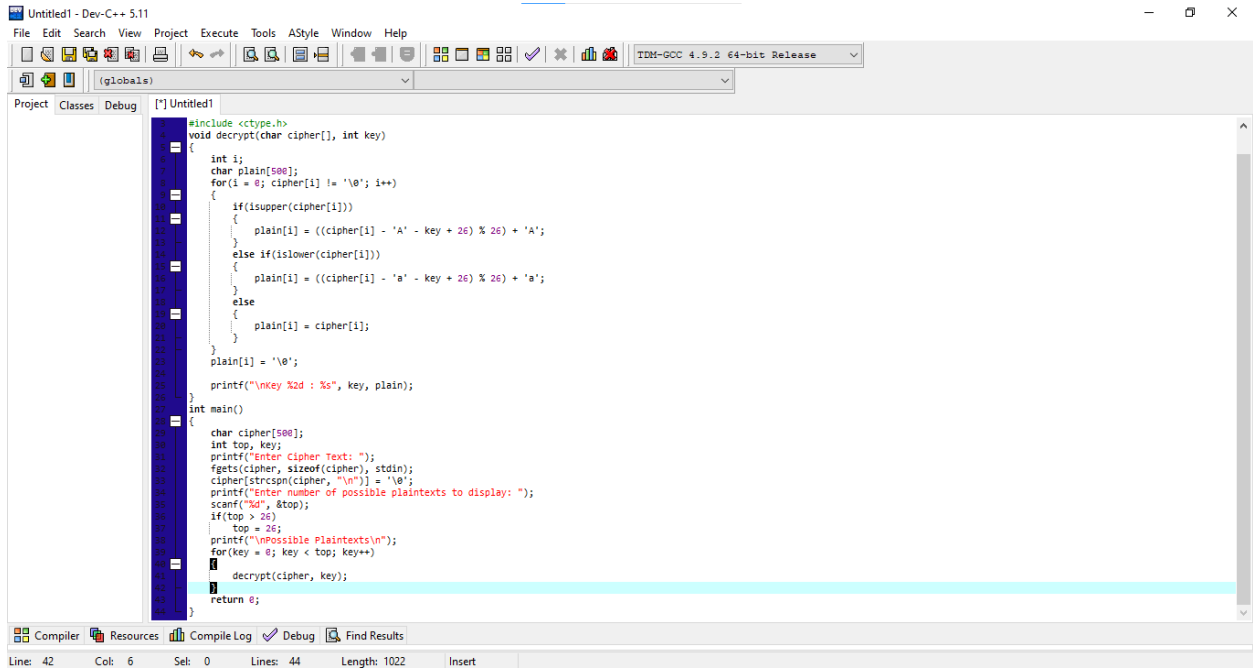
Algorithm:

1. Start the program.
2. Accept the ciphertext from the user.
3. Accept the number of possible plaintexts (Top N) to display.
4. For each possible key from **0 to 25**, decrypt the ciphertext using the additive cipher formula:
 - $P = (C - K + 26) \bmod 26$
5. Preserve spaces and non-alphabetic characters during decryption.
6. Store the decrypted plaintext corresponding to each key.
7. Display the decrypted plaintexts in order of keys.
8. Print only the first **Top N** possible plaintexts specified by the user.
9. Stop the program.

Result:

The program successfully performs a **letter frequency attack (brute-force attack)** on an **Additive (Caesar) Cipher** by generating possible plaintexts for different key values. It displays the **Top N** candidate plaintexts specified by the user, allowing the most meaningful plaintext to be identified. This demonstrates how the small key space of the Caesar cipher makes it vulnerable to cryptanalysis.

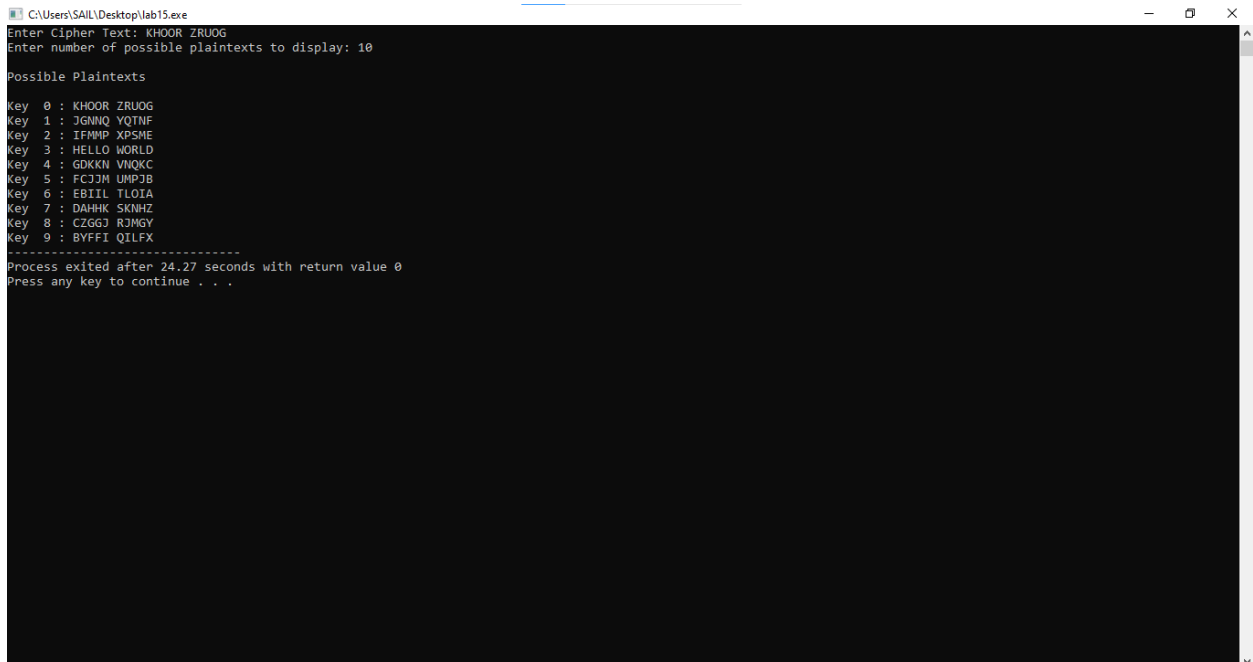
Code:



```
#include <ctype.h>
void decrypt(char cipher[], int key)
{
    int i;
    char plain[500];
    for(i = 0; cipher[i] != '\0'; i++)
    {
        if(isupper(cipher[i]))
        {
            plain[i] = ((cipher[i] - 'A' - key + 26) % 26) + 'A';
        }
        else if(islower(cipher[i]))
        {
            plain[i] = ((cipher[i] - 'a' - key + 26) % 26) + 'a';
        }
        else
        {
            plain[i] = cipher[i];
        }
    }
    plain[i] = '\0';
    printf("\nKey %2d : %s", key, plain);
}

int main()
{
    char cipher[500];
    int top, key;
    printf("Enter Cipher Text: ");
    fgets(cipher, sizeof(cipher), stdin);
    cipher[strlen(cipher) - 1] = '\0';
    printf("Enter number of possible plaintexts to display: ");
    scanf("%d", &top);
    if(top > 26)
        top = 26;
    printf("\nPossible Plaintexts\n");
    for(key = 0; key < top; key++)
    {
        decrypt(cipher, key);
    }
    return 0;
}
```

Output:



```
C:\Users\SAIL\Desktop\lab15.exe
Enter Cipher Text: KHOOR ZRUOG
Enter number of possible plaintexts to display: 10

Possible Plaintexts

Key 0 : KHOOR ZRUOG
Key 1 : JGNMQ YQTNF
Key 2 : IFMMP XPSME
Key 3 : HELLO WORLD
Key 4 : GDKKN VNQKC
Key 5 : FCJJM UMPJB
Key 6 : EBIIJ TLOJA
Key 7 : DAHHK SKWHZ
Key 8 : CZGGJ RJNGY
Key 9 : BVFFI QILFX
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Process exited after 24.27 seconds with return value 0
Press any key to continue . . .
```